

3). toxicity filtering: using rules and jigsaw classifier

4). Deduplications using Bloom filtering

Result: 57 tokens

pro Model-based filtering:

positive examples (200k): open/termes - 2.5: mostly GPT-4 ^{interaction data}
negative examples (200k): { 2 (i) 5: subreddit with curiosity
Q/A
Refinedweb.

Result: 3.87 tokens

Trained a classifier, ran it on all of DCLM-pool, this
fastest quality classifier outperforms other filtering
methods

Ref-nemotron-cc(): from Nvidia.

Motivation: FinewebEdu and DCLM datasets filtered too aggressively
(or remove 90% of data)

gaps { 2: Need more tokens (but preserve quality)

3: For HTML $\rightarrow$ text, used just text (because) it returned
more tokens

Solutions:

1. classifier ensembling:

-- prompt Nemotron-340B-instruct to score Fineweb
documents based on educational value, disill into
-- DCLM classifier faster model

②: Synthetic data rephrasing:

- -- for low-quality data, use LM to rephrase low-quality data
- -- for high-quality data, use LM to generate easier data (QA pairs, extract key information)

Result: 6.3T tokens

Llama3 trained on 15T / Gwens trained on 36T

copy-right (Lawsuits): Lots of ~~copy-right~~ around generative AI, mostly around "copyright"

copyright law: incentive creation goods: paper, patent, codes
most things on the Internet are actually copyrighted

How to copyright your work?

⇒ Licenses

⇒ Apple ^{appeal to} ~~Apple~~ Fair Use

def long-context():

Demand for long contexts (out on books)

def tasks(): Convert lots of existing NLP datasets into prompts

def instruction-chat():

more open-ended instructions
heavy use of synthetic data

Terms of service(): privacy / prohibit downloading videos

Summary:

⇒ Key lesson: Data does fall from the sky, you have to get it

⇒ live service ⇒ raw data ⇒ processed data (conversion, filtering, deduplication)

• ⇒ legal and ethical issues (e.g. copyright and privacy)

⇒ much of this pipeline is heuristic, many opportunity to improve